

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using content-based filtering approach	
Experiment No: 11	Date:	Enrolment No: 92410133023

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.

In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:

In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.

Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:

The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

Advantages and Disadvantages:

- **Advantages:**
 - No need for data on other users when applying to similar users.
 - Able to recommend to users with unique tastes.
 - Able to recommend new & popular items
 - Explanations for recommended items.

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- **Disadvantages:**

- Finding the appropriate feature is hard.
- Doesn't recommend items outside the user profile.

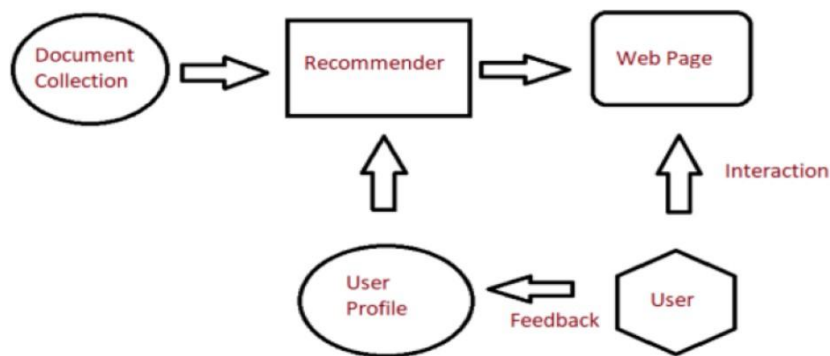


Fig: Recommender System

Pre Lab Exercise:

1. When can you use content-based recommendation system approach?

Ans: it is used when item features are available and user preferences are known, useful when there is no data about other users, works well for personalized recommendations, best for systems like movies, news, and products.

2. Write advantages of content-based filtering approach

Ans: does not require data from other users, can recommend new or unseen items, works well for users with unique preferences, provides explainable recommendations.

3. Write disadvantages of content-based filtering approach

Ans: limited to user past interests, requires good features extraction, cannot recommend completely new types of items, may lead to over specialization.

Program (Code):

```
#lab 11
```

```
# Step 1: Import Libraries
```

```
import pandas as pd
```

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```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

Step 2: Sample Dataset

```
data = {
'title': ['Movie A', 'Movie B', 'Movie C', 'Movie D'],
'description': [
'action adventure hero fight',
'romantic love story drama',
'action thriller suspense',
'comedy fun entertainment'
]
}
```

```
df = pd.DataFrame(data)
```

Step 3: TF-IDF Vectorization

```
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(df['description'])
```

Step 4: Similarity Matrix

```
similarity = cosine_similarity(tfidf_matrix)
```

Step 5: Recommendation Function

```
def recommend(movie):
idx = df[df['title'] == movie].index[0]
scores = list(enumerate(similarity[idx]))
scores = sorted(scores, key=lambda x: x[1], reverse=True)
print("Recommended Movies:")
for i in scores[1:4]:
print(df.iloc[i[0]]['title'])
```

Step 6: Test

```
recommend("Movie A")
```

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Results:

```
Recommended Movies:
Movie C
Movie B
Movie D
```

Observation: the system recommends items based on similarity of content features, TF-IDF helps in identifying important words in item descriptions, cosine similarity measures how closely items are related, recommendations are accurate for similar content, it may lack diversity in suggestions.

Post Lab Exercise:

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.

Ans: the system can be improved by adding more features like genre, ratings, and user feedback, hybrid recommendation can improve accuracy, using deep learning embeddings can enhance semantic understanding, personalization can be improved using user interaction history.

2. Recommend top-10 "series" using IMDB movie dataset
(Link: <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>). Attach the code with output. Also, write your observation for the same.

Code:

```
# Step 1: Import Libraries
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Step 2: Load Dataset
df = pd.read_csv('IMDB-Movie-Data.csv')

# Step 3: Check Columns (optional)
print(df.columns)
```

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```
# Step 4: Use required columns
# This dataset uses 'Title' and 'Genre'
df = df[['Title', 'Genre']]
df = df.dropna()
```

```
# Step 5: TF-IDF Vectorization on Genre
tfidf = TfidfVectorizer()
tfidf_matrix = tfidf.fit_transform(df['Genre'])
```

```
# Step 6: Cosine Similarity
similarity = cosine_similarity(tfidf_matrix)
```

```
# Step 7: Recommendation Function
def recommend(movie_name):
    if movie_name not in df['Title'].values:
        print("Movie not found!")
        return
    idx = df[df['Title'] == movie_name].index[0]
    scores = list(enumerate(similarity[idx]))
    scores = sorted(scores, key=lambda x: x[1], reverse=True)
```

```
print("\nTop 10 Recommendations:\n")
count = 0
for i in scores:
    if i[0] != idx:
        print(df.iloc[i[0]]['Title'])
        count += 1
    if count == 10:
        break
```

```
# Step 8: Test
recommend("Guardians of the Galaxy")
```

Output:

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```
Index(['Rank', 'Title', 'Genre', 'Description', 'Director', 'Actors', 'Year',
      'Runtime (Minutes)', 'Rating', 'Votes', 'Revenue (Millions)',
      'Metascore'],
      dtype='object')
```

Top 10 Recommendations:

```
Rogue One
Independence Day: Resurgence
X-Men: Apocalypse
Captain America: Civil War
Star Trek Beyond
Batman v Superman: Dawn of Justice
Mad Max: Fury Road
Inception
Jurassic World
Avengers: Age of Ultron
```

The system successfully recommends similar series based on genre, TF-IDF captures important genre features for similarity calculation, recommendations are relevant but limited to genre similarity only, some recommendations may not match user preferences exactly, system performances can improve with more features and hybrid methods.